Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE306	COMPUTER NETWORKS	3	0	0	3	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					

Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT

Course Outcomes :Through this course students should be able to

CO1 :: Recognize various network hardware component and their topologies

CO2 :: Explain the concepts of data communication modulation and multiplexing in physical layer

CO3 :: Apply principles of data link layer design and employ multiple access protocols

CO4 :: Utilize IP addressing and packet forwarding technique in network setups

CO5 :: Examine different routing algorithm and the performance of unicast routing protocols in network environment

CO6 :: Classify the role and functioning of key application layer protocols and services

It is mandatory to complete the number of courses for being eligible for End Term Examination along with the attendance criteria of the university. The links of the courses as shared in the IP should be completed on/before the last teaching day as per the academic calendar of the university.

Three evaluations will be conducted for each CA on OAS platform. Each evaluation will consist of 10 MCQ type questions, with each question carrying 1 mark, making a total of 10 marks per evaluation. Hence, the total marks for one CA will be 30. Two best CAs out of three will be considered.

Note 1:- No Separate CA shall be conducted.

Note 2: - The marks obtained in CAs will be further prorated based on the percentage of course(s) completion on the Coursera platform.

Note 3:- Red flags will be raised if any discrepancies are detected, such as skipping videos, increasing video playback speed, or prolonged periods of inactivity. In such cases, the completion certificate will not be eligible for mapping to this course and student will not be allowed to appear for End Term Exam.

Relevant Websites (RW)		
Sr No	**Web address**	**Course Name**
RW-1	https://www.coursera.org/learn/computer-networking	The Bits and Bytes of Computer Networking
RW-2	https://www.coursera.org/learn/fundamentals-network-communications?specialization=computer-communications	Fundamentals of Network Communication
RW-3	https://www.coursera.org/learn/peer-to-peer-protocols-local-area-networks?specialization=computer-communications	Peer-to-Peer Protocols and Local Area Networks
RW-4	https://www.coursera.org/learn/packet-switching-networks-algorithms?specialization=computer-communications	Packet Switching Networks and Algorithms
RW-5	https://www.coursera.org/learn/tcp-ip-advanced?specialization=computer-communications	TCP/IP and Advanced Topics

LTP week distribution: (LTP Weeks)

Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan for Lectures

Week Number	Tentative Date	Broad Topic	Sub Topics	Link of Sub Topics
Week 1	12/8/2024-18/8/2024	INTRODUCTION	INTRODUCTION(Uses of Computer Networks)	https://www.coursera.org/learn/computer-networking/lecture/7WFyg/course-introduction
	12/8/2024-18/8/2024		Networks and Types	
	12/8/2024-18/8/2024		Introduction to internet, browser, web server, URL, domain name, Ip address and Packets	https://www.coursera.org/learn/fundamentals-network-communications?specialization=computer-communications
	12/8/2024-18/8/2024		Network hardware architecture and its topologies and device like HUB, Switch and Routers	https://www.coursera.org/learn/computer-networking/lecture/7WFyg/course-introduction
Week 2	19/8/2024-25/8/2024	NETWORK MODELS	Network software architecture and its layers and protocols	https://www.coursera.org/lecture/fundamentals-network-communications/osi-unified-view-of-protocols-and-services-DRUcN
	19/8/2024-25/8/2024		NETWORK MODELS (Protocol Layering)	
	19/8/2024-25/8/2024		NETWORK MODELS (TCP/IP protocol suite)	https://www.coursera.org/lecture/fundamentals-network-communications/tcp-ip-architecture-and-routing-examples-RJ6pg
	19/8/2024-25/8/2024		NETWORK MODELS(OSI Model)	https://www.coursera.org/lecture/fundamentals-network-communications/layered-architecture-and-osi-model-njImK
Week 3	26/8/2024-1/9/2024		NETWORK MODELS (TCP/IP protocol suite)	https://www.coursera.org/lecture/fundamentals-network-communications/tcp-ip-architecture-and-routing-examples-RJ6pg
	26/8/2024-1/9/2024		NETWORK MODELS(OSI Model)	https://www.coursera.org/lecture/fundamentals-network-communications/layered-architecture-and-osi-model-njImK
	26/8/2024-01/9/2024	PHYSICAL LAYER: Signal & Media	PHYSICAL LAYER: Signal & Media(Basics for Data Communications and Analog and Digital signals)	Textbook
	26/8/2024-01/9/2024		PHYSICAL LAYER: Signal & Media(Data Rate)	
	26/8/2024-01/9/2024		PHYSICAL LAYER: Signal & Media(Basics for Data Communications and Analog and Digital signals)	
	26/8/2024-01/9/2024		PHYSICAL LAYER: Signal & Media(Data Rate)	

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 4	02/9/2024-08/9/2024	PHYSICAL LAYER: Conversions	PHYSICAL LAYER: Signal & Media(Transmission Impairments and Performance)	https://www.coursera.org/videos/computer-networking/Nihjd?query=physical+layer+&source=search
	02/9/2024-08/9/2024		PHYSICAL LAYER: Signal & Media(Transmission media like Guided and Unguided media)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Signal & Media(Cabling standards)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Digital to Digital Conversion)	Textbook
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Analog to Digital Conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Analog to Analog conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Digital to Analog conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Digital to Digital Conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Analog to Digital Conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Analog to Analog conversion)	
	02/9/2024-08/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Digital to Analog conversion)	
Week 5	09/9/2024-15/9/2024	PHYSICAL LAYER: Modulation & Multiplexing	PHYSICAL LAYER: Modulation & Multiplexing (Digital to Digital Conversion)	
	09/9/2024-15/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Analog to Digital Conversion)	
	09/9/2024-15/9/2024		PHYSICAL LAYER: Modulation & Multiplexing	

			(Analog to Analog conversion)	
	09/9/2024-15/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Digital to Analog conversion)	
	09/9/2024-15/9/2024		PHYSICAL LAYER: Modulation & Multiplexing (Multiplexing)	Textbook
	Test 1			
Week 6	16/9/2024-22/9/2024	DATA LINK LAYER	DATA LINK LAYER(Data link Layer design issues)	
	16/9/2024-22/9/2024		DATA LINK LAYER(Error Detection and Correction, Hamming code, CRC, Parity, Checksum)	https://www.coursera.org/learn/fundamentals-network-communications/lecture/BS6X4/error-control-parity-checks
	16/9/2024-22/9/2024		DATA LINK LAYER (Switch working)	
	16/9/2024-22/9/2024		DATA LINK LAYER (Elementary Data link Protocols)	
	16/9/2024-22/9/2024		DATA LINK LAYER (Elementary Data link Protocols)	
Week 7	23/9/2024-29/9/2024	MAC Sub Layer	MAC SUBLAYER(Multiple Access Protocols- ALOHA, CSMA and CSMA/CD)	https://www.coursera.org/learn/peer-to-peer-protocols-local-area-networks?specialization=computer-communications
	23/9/2024-29/9/2024		MAC SUBLAYER(Random Access)	
	23/9/2024-29/9/2024		MAC SUBLAYER (Controlled access)	
	23/9/2024-29/9/2024		MAC SUBLAYER(Ethernet protocol)	
	23/9/2024-29/9/2024		Spill Over	
	MTE			
Week 8	07/10/2024-13/10/2024	NETWORK LAYER	NETWORK LAYER: IP Addressing(Network layer design issue)	https://www.coursera.org/learn/computer-networking/lecture/AexeX/the-network-layer
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(Network layer	

			services)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(IP Addressing Both Classfull and Classless)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(Network layer performance)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(Forwarding of IP packets)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(Subnetting and Supernetting)	https://www.coursera.org/learn/computer-networking/home/module/2
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(Subnetting examples)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(IP Header)	
	07/10/2024-13/10/2024		NETWORK LAYER: IP Addressing(IPv6 addressing)	
Week 9	14/10/2024-20/10/2024		NETWORK LAYER: IP Addressing(Subnetting and Supernetting)	https://www.coursera.org/learn/computer-networking/home/module/2
	14/10/2024-20/10/2024		NETWORK LAYER: IP Addressing(Subnetting examples)	
	14/10/2024-20/10/2024		NETWORK LAYER: IP Addressing(IP Header)	
	14/10/2024-20/10/2024		NETWORK LAYER: IP Addressing(IPv6 addressing)	
	14/10/2024-20/10/2024		NETWORK LAYER: IP Addressing(IPv6 addressing)	
	14/10/2024-20/10/2024		NETWORK LAYER: Routing(Routing algorithms)	
	14/10/2024-20/10/2024		NETWORK LAYER: Routing(Unicast routing protocols)	https://www.coursera.org/programs/lpu-faculty-development-program-f9ei7/learn/packet-switching-networks-algorithms?fromClip=sfc_page_course_link~Aw3SY
Week 10	21/10/2024-27/10/2024	Network Layer (Routing)	NETWORK LAYER: Routing(Routing algorithms)	

			NETWORK LAYER: Routing(Unicast routing protocols)	
	Test 2			
	21/10/2024-27/10/2024	NETWORK LAYER(Congestion Control)	NETWORK LAYER: Routing(Routing Algorithm Shortest path algorithm)	https://www.coursera.org/learn/packet-switching-networks-algorithms/home/module/3
	21/10/2024-27/10/2024		NETWORK LAYER: Routing(Distance vector Routing)	
	21/10/2024-27/10/2024		NETWORK LAYER: Routing(Link State routing)	
Week 11	28/10/2024-03/11/2024		NETWORK LAYER: Congestion Control (Congestion Control Algorithms)	
	28/10/2024-03/11/2024		NETWORK LAYER: Congestion Control (Congestion Control Algorithms)	
	28/10/2024-03/11/2024		TRANSPORT LAYER (Transport Layer Services)	https://www.coursera.org/learn/computer-networking/lecture/cboIM/the-transport-layer
	28/10/2024-03/11/2024		TRANSPORT LAYER (TCP-Header format and handshaking operation)	
	28/10/2024-03/11/2024		TRANSPORT LAYER (UDP-Header format)	
Week 12	04/11/2024-10/11/2024	TRANSPORT LAYER	TRANSPORT LAYER (Transport Layer Services)	
	04/11/2024-10/11/2024		TRANSPORT LAYER (TCP-Header format and handshaking operation)	
	04/11/2024-10/11/2024		TRANSPORT LAYER (UDP-Header format)	
	Test 3			
	04/11/2024-10/11/2024	APPLICATION LAYER	APPLICATION LAYER(E Mail)	https://www.coursera.org/learn/computer-networking/lecture/aZPjv/why-do-we-need-dns
Week 13	11/11/2024-17/11/2024		APPLICATION LAYER(E Mail)	
	11/11/2024-17/11/2024		APPLICATION LAYER (Domain Name System)	

	11/11/2024-17/11/2024		APPLICATION LAYER (Domain Name System)	
	11/11/2024-17/11/2024		APPLICATION LAYER (FTP)	
Week 14	11/11/2024-17/11/2024		Spill Over	
Week 15	18/11/2024-24/11/2024		Spill Over	
	18/11/2024-24/11/2024		Spill Over	
	18/11/2024-24/11/2024		Spill Over	

Scheme for CA:

CA Category of this Course Code is: A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test 1	50	CO1, CO2
Test 2	50	CO3, CO4
Test 3	50	CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allotment / submission Week
Test 1	To check the learning level	MCQ Based Class Test	Individual	Online	30	3/5
Test 2	To Check the Learning level	MCQ Based Class Test	Individual	Online	30	7/9

Test 3	To check the learning level	MCQ Based Class Test	Individual	Online	30	11/13